

ROBERT VOKÁČ, BSc

C++ Systems Engineer (Runtime / Compatibility Layers)

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PROFILE

C++ engineer focused on **low-level systems, runtime design, and API compatibility layers**.

Designed and implemented **custom runtimes replacing XNA, DirectX 3, and WinAPI**, enabling real-world games to run without original dependencies across Windows, Linux, Web, and Android.

Experience includes **reverse-engineering behavior, reproducing runtime semantics, and building cross-platform abstractions over SDL3/OpenGL**.

Transitioning from Java backend development to **full-time C++ systems engineering**, with focus on runtime and compatibility layers.

Seeking roles in **systems engineering, runtime/framework development, game technology, or backend/platform architecture**.

CORE PROJECTS

CNA

GitHub: github.com/openeggbert/cna

Custom cross-platform C++ runtime reimplementing the XNA 4.0 API over SDL3 (~5k LOC, production-tested on real game across Windows, Linux, Web, and Android)

- Designed an **XNA-compatible runtime layer** matching original game-loop, input, rendering, audio, and resource-lifecycle behavior
- Implemented **rendering abstraction** supporting SDL_Renderer, OpenGL, and bgfx backends
- Ported real-world game (**Speedy Blupi, 2013**) from C# / XNA to C++
- Achieved working, smooth builds on Windows, Linux, Web, and Android from the same C++/CNA codebase
- Replaced **MonoGame/.NET/XNA** dependencies with a native **C++/SDL3-based architecture**
- Reproduced original runtime behavior **without access to original engine internals**
- Achieved **smooth runtime performance (~0.1% CPU usage, ~47 MB RAM)**

Playable web build: <https://speedyblupi.com/SpeedyBlupi2013/>

Video demo: <https://www.youtube.com/watch?v=TScl1L-dbHE>

Free Direct + Free API

GitHub: github.com/openeggbert/free-direct · github.com/openeggbert/free-api

Custom reimplementation of DirectX 3 (2D) and WinAPI subset in modern C++ over SDL3 (~3k LOC per module)

- Implemented **DirectDraw-style 2D rendering and WinAPI-style event/input handling**.

- Built compatibility layer enabling execution of **legacy Windows applications without original dependencies**.
- Supports execution of:
 - **Speedy Blupi (1998/2001)** without DirectX SDK or WinAPI
 - **Planet Blupi (1996)** (partial compatibility, ongoing debugging)

Playable web build: <https://speedyblupi.com/SpeedyEggbert2/>

Playable web build: <https://speedyblupi.com/PlanetBlupi/>

Video demo: <https://www.youtube.com/watch?v=TI3vNrqqgtbo>

Hive (archived)

GitHub: github.com/robertvokac/hive

40k+ LOC modular backend platform

- Designed **metadata-driven architecture, custom ORM, and REST generation**
- Implemented **plugin system, scheduler, triggers, and validation pipeline**

SKILLS

Languages: C++20/23 (primary), SQL

Systems & Tools: SDL3, OpenGL, CMake, Linux, SQLite, Git

Core Areas:

- Runtime and framework architecture
- API compatibility layers (XNA / DirectX / WinAPI)
- Rendering and system integration
- Modular systems and abstraction design
- Debugging complex systems and legacy behavior

EXPERIENCE

Software Engineer – Assist (2017 - present)

- Maintained and improved **production backend systems (Java, SQL)**
- Diagnosed and resolved **production issues, improving system stability**
- Applied **system design principles in long-lived codebases**
- Developed **C++ systems independently (runtime and compatibility layers)**

Technologies: Java, SQL, Git, Bash

EDUCATION

BSc in Computer Science

Czech University of Life Sciences Prague (2014-2017)

LANGUAGES

English - B2

Czech - native